EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 14

Assignment: Integrating OOAD Models into Design

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 14 of 16

Contents

ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 14 module and your lecture notes before attempting each question.
- Use the university course registration case study for your examples.
- Design classes must show visibility, data types, parameters and return types.
- Show how your models are consistent: messages must match operations.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define object-oriented design and differentiate it from object-oriented analysis. [3 marks]

Question A2

Explain why analysis models must be transformed into design models. [2 marks]

Question A3

Define a design class and list four details it adds over an analysis class. [3 marks]

Question A4

What is visibility and why is it important in design classes? [2 marks]

Question A5

Define a parameter and explain its role in design operations. [2 marks]

Question A6

Explain the purpose of return types in design. [2 marks]

Question A7

What is a design pattern and how does it support good design? [2 marks]

Question A8

Explain how object, dynamic and behavioural models integrate in design. [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

Consider integrating the Object, Dynamic and Behavioural models for a registration system.

- (a) Show how the class diagram provides structure. [3 marks]
- (b) Show how the state diagram adds state behaviour. [3 marks]
- (c) Show how the sequence diagram adds interaction detail. [3 marks]

- (d) Explain how these models together provide a complete design. [3 marks]

Question B2 (8 marks)

Consider a Student class with analysis attributes and operations.

- (a) Transform it into a design class with visibility and types. [4 marks]
- (b) Explain how this transformation supports implementation. [4 marks]

Section C — Case Study (10 marks)

Consider the complete university management system with all three models.

As an object-oriented designer:

- (a) Explain how to integrate all three models into a cohesive design. [4 marks]
- (b) Identify potential inconsistencies and how to resolve them. [3 marks]
- (c) Explain the value of integrated modelling for implementation. [3 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (8 questions)	20
B	Application questions (12 + 8)	20
C	Case study — complete system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Ensure your design models are complete and consistent.
- Cite any sources you consult using the format used in the Week 14 module references.